

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA0706 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Dr.V.PARTHIPAN

Code of Conduct:

I __Maniyam Anushka(192525411)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SIMATS ENGINEERING

ASSESSMENT TOOL 2 — ANSWER SCRIPT

Course Name: CSA07 Course Code: Computer Networks

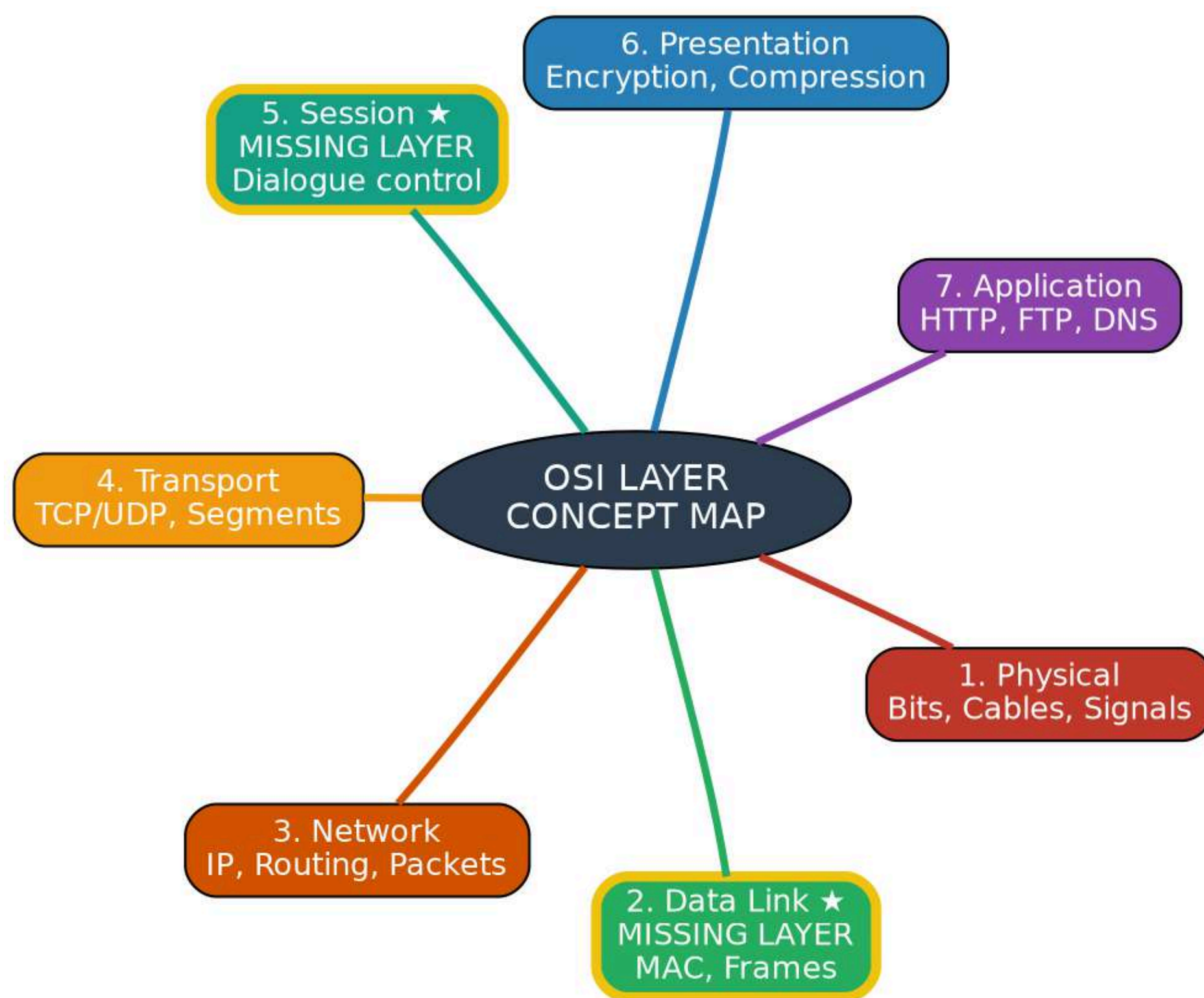
Course Outcome: CO1 — Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Annexure A — Concept Mapping

Question 1 — Completing the Concept Map of OSI Layers

Application → Presentation → _____ → Transport → Network → _____ → Physical.

The two missing layers are the Session Layer (between Presentation and Transport) and the Data Link Layer (between Network and Physical). The complete concept map is:



Why these layers are essential:

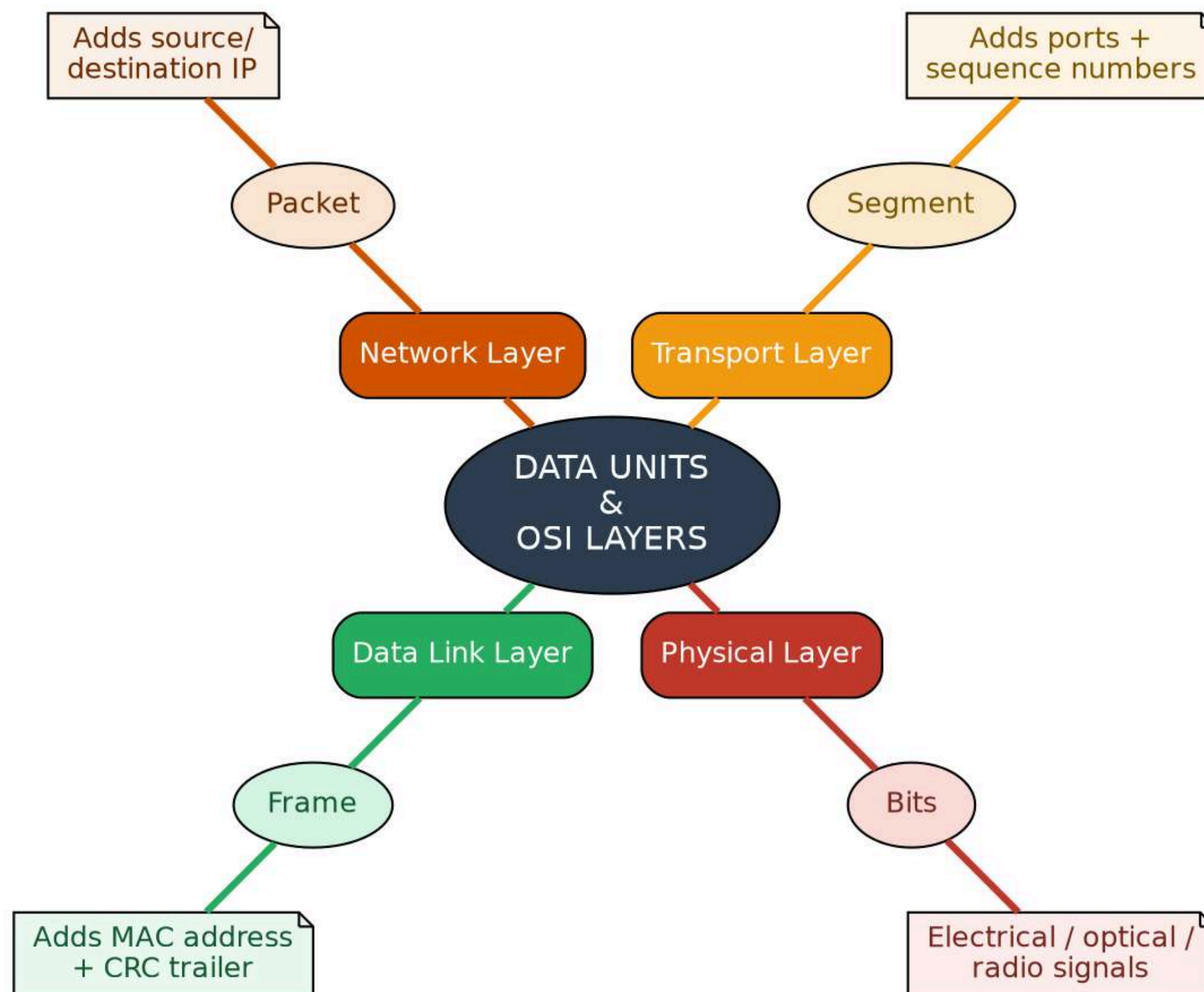
- Session Layer: Establishes, manages, and terminates the connection (session) between two communicating applications. It handles synchronization and dialogue control, ensuring that if a session is interrupted, communication can resume from a checkpoint rather than restarting — this supports reliability during long transfers.***
- Data Link Layer: Provides node-to-node delivery on the same physical network using MAC addressing, frames the data received from the Network layer, and performs error detection (via CRC) so that corrupted frames are identified before being passed up the stack.***

Together with the other five layers, these two layers complete the chain of encapsulation and control needed for end-to-end reliable communication: without a Session layer, applications

would lose track of ongoing dialogues, and without a Data Link layer, raw bits could not be reliably framed and addressed on the local network.

Question 2 — Data Units and Their OSI Layers

Concept map linking each Protocol Data Unit (PDU) to its corresponding OSI layer:



Analysis — how transformation of data units supports transmission:

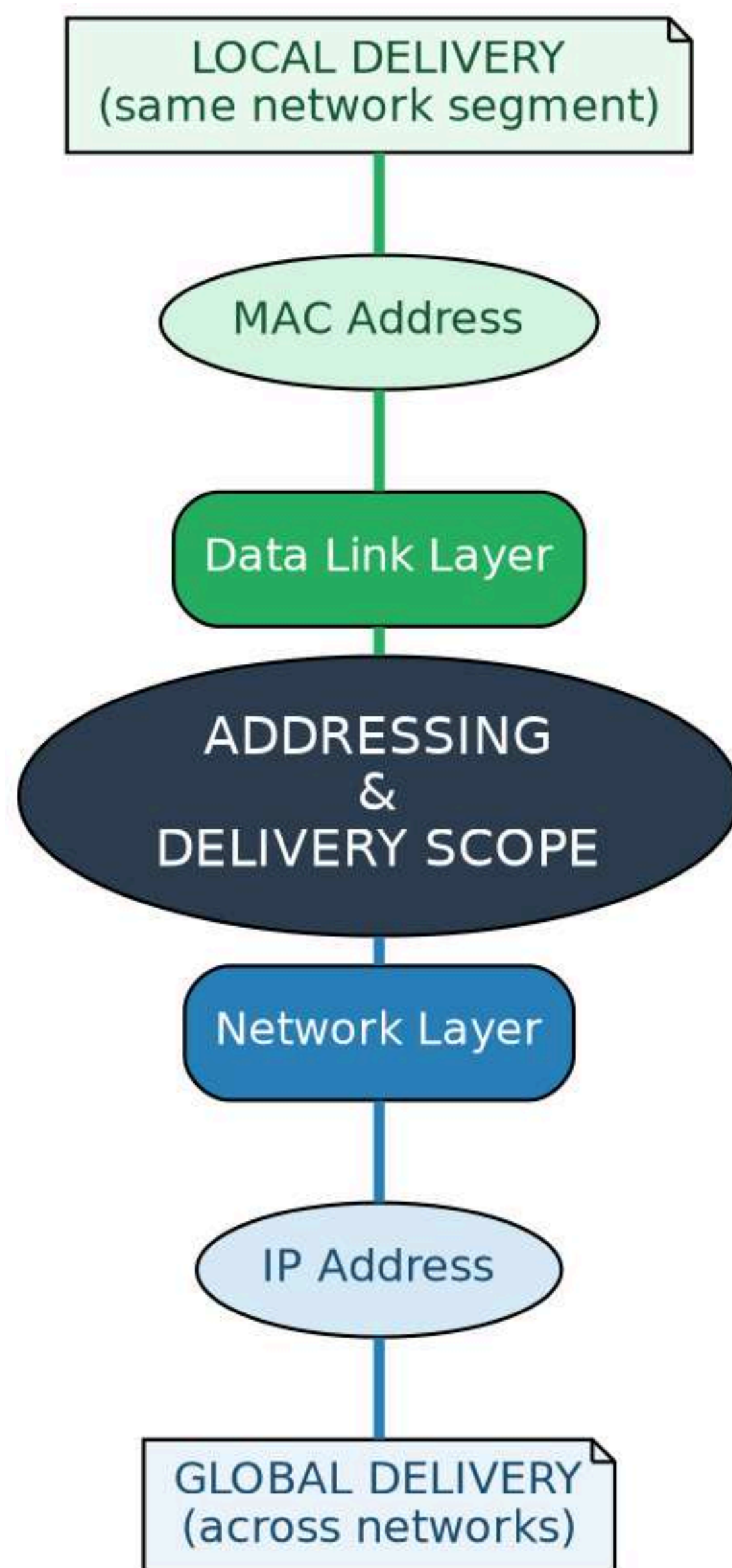
As data travels down the sender's protocol stack, each layer encapsulates the unit received from the layer above by adding its own header (and, at the Data Link layer, a trailer). This is called encapsulation: Data → Segment → Packet → Frame → Bits. At the receiver, the reverse process, decapsulation, strips each header layer by layer, moving Bits → Frame → Packet → Segment → Data. This transformation ensures that each layer only needs to understand its own header, allowing addressing, routing, and error checking to happen independently at each stage — which makes the overall transmission modular, reliable, and easy to troubleshoot.

Question 3 — MAC Address vs IP Address in Data Delivery

Given concept map:

Data Link → MAC Address → Local Delivery

Network → IP Address → Global Delivery



Analysis:

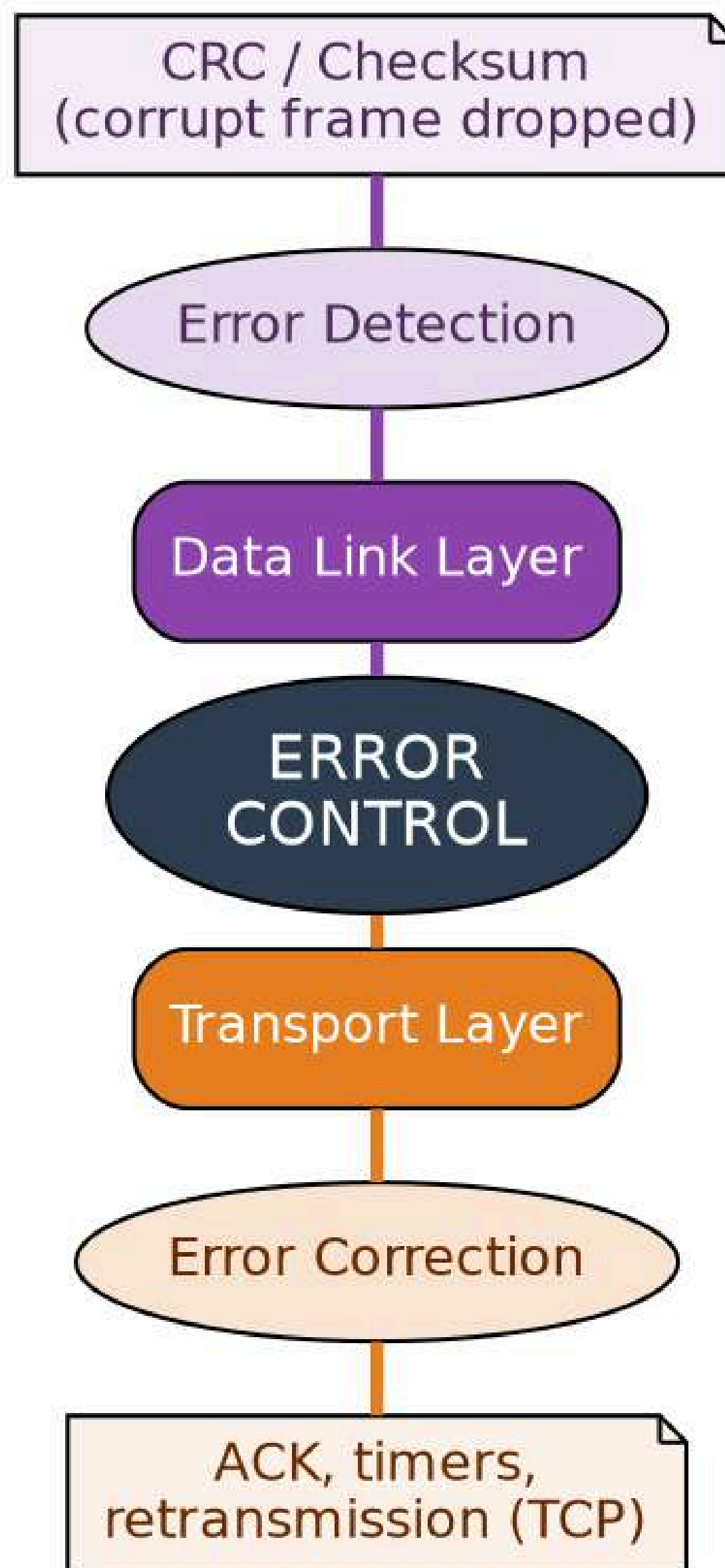
These two addressing schemes work together to achieve complete, end-to-end delivery. The IP address identifies the ultimate source and destination hosts globally, guiding the packet across multiple networks hop by hop (Network layer / global delivery). At every hop, the MAC address identifies the specific next-hop device on the local segment, so the Data Link layer can hand the frame to the correct physical neighbor (local delivery). A router examines the destination IP address to decide the next hop, then re-encapsulates the packet in a new frame addressed to that next hop's MAC address. This layered addressing — logical for the whole journey, physical for each local step — ensures data reaches the correct destination across both local networks and the wider internet.

Question 4 — Error Control at Data Link vs Transport Layer

Given concept map:

Error Control → Data Link Layer → Error Detection

Error Control → Transport Layer → Error Correction

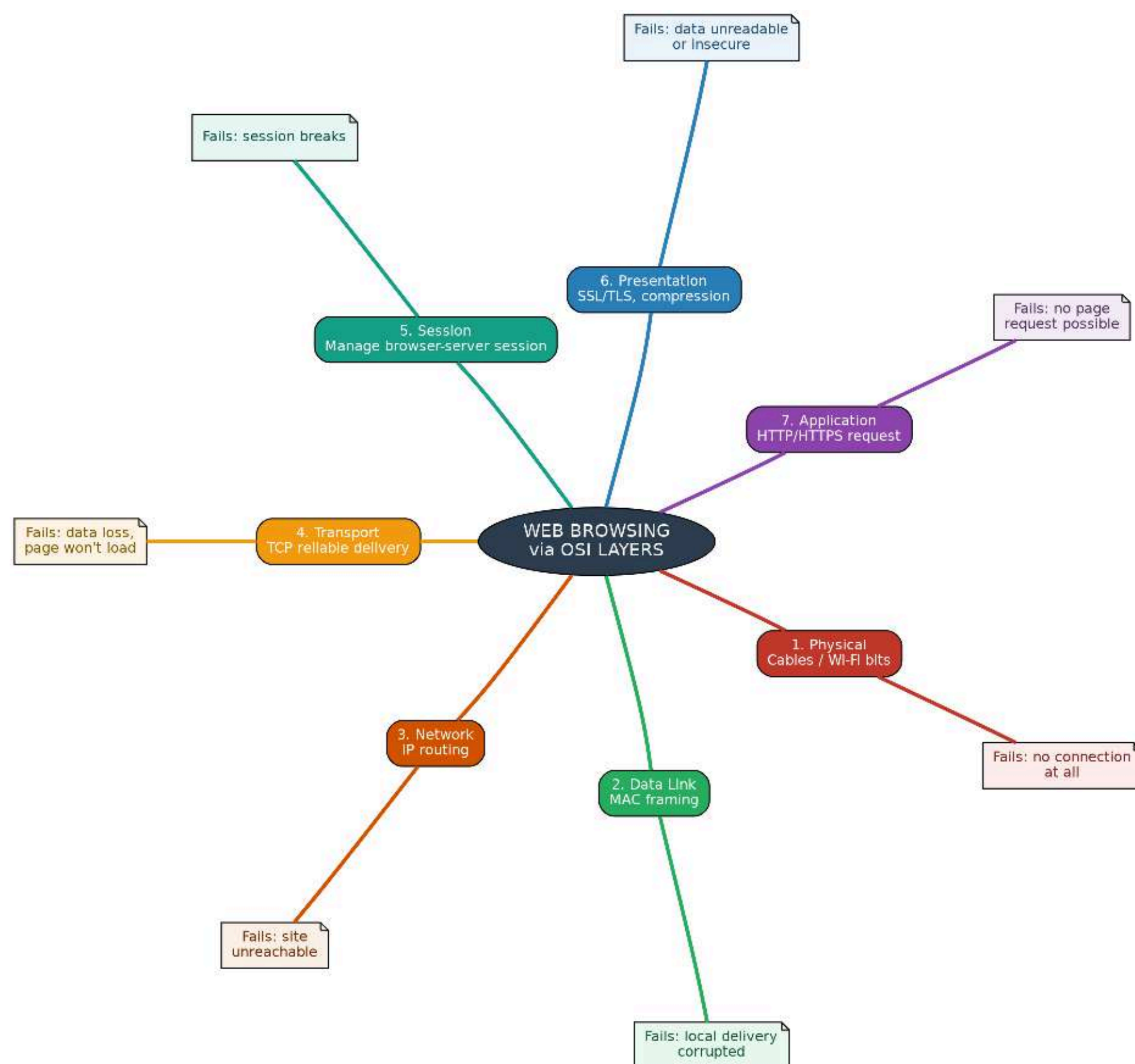


Analysis — interaction between the layers:

The Data Link layer acts as the first line of defense: it checks each frame locally, hop by hop, and simply drops any frame found to be corrupted — it does not attempt to fix it. This prevents damaged data from propagating further into the network. The Transport layer then provides the end-to-end safety net: because it does not know why a segment failed to arrive (whether due to a Data Link error, congestion, or a routing issue), it uses acknowledgements and timeouts to detect the loss and retransmit the missing segment. This two-tier strategy — fast, local detection at Layer 2 and reliable, end-to-end correction at Layer 4 — improves overall reliability without requiring every intermediate device to perform expensive error-recovery work.

Question 5 — Concept Map for Web Browsing Using OSI Layers

User types a URL in the browser → the request flows down through all seven OSI layers, is transmitted, and the response flows back up through the same layers on the client side.

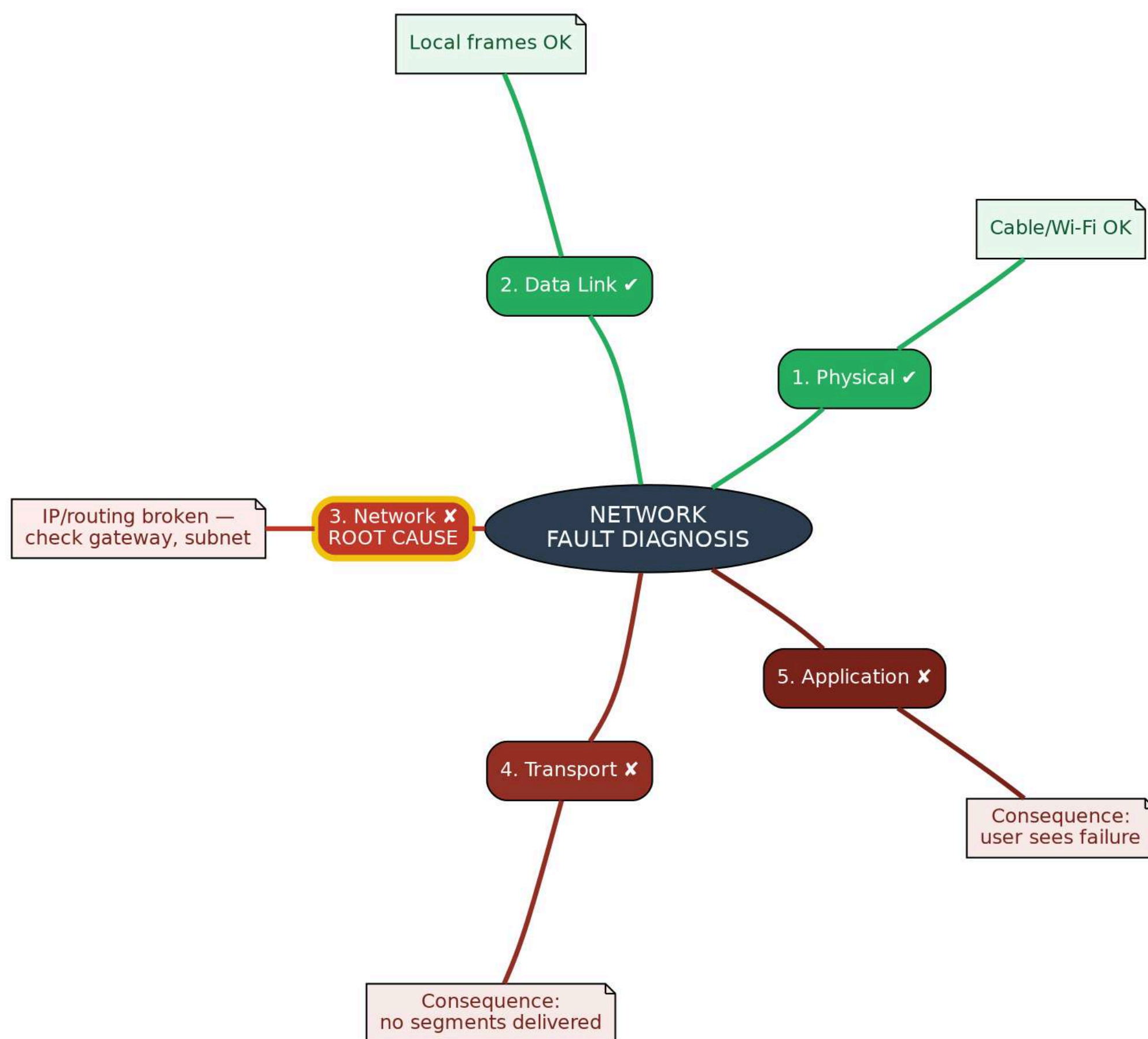


Analysis:

All seven layers work together in sequence — each layer depends on the correct functioning of the one below it. When a user browses the web, the request is encapsulated downward from Application to Physical at the client, transmitted, and de-encapsulated upward from Physical to Application at the server (and again for the response). Because the layers are interdependent, a failure at any single layer breaks the entire chain: for instance, even if the Application, Presentation, and Session layers work perfectly, a Physical layer failure (a disconnected cable or lost Wi-Fi signal) means no bits can be transmitted, so no web page can ever load. This illustrates why reliable web browsing requires every OSI layer to function correctly.

Question 6 — Troubleshooting Using the Concept Map

Given status: Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗



Analysis — root cause and value of the concept map:

The pattern of ticks and crosses shows that the problem originates at the Network layer: everything below it (Physical and Data Link) is confirmed working, so the fault is not a cabling or local-network issue. Everything above it (Transport and Application) fails only as a downstream consequence — those layers have nothing to build on once packets stop being routed correctly. Likely causes at the Network layer include an incorrect IP address/subnet configuration, a missing or misconfigured default gateway, or a routing failure. The concept map makes troubleshooting efficient by allowing a technician to work bottom-up: as soon as a layer is confirmed working while the next one fails, the search for the root cause can be narrowed to that specific layer, avoiding wasted effort investigating layers that are already known to be fine.